MILITARY SPECIFICATION SHEET

PLASTIC SHEET, LAMINATED, METAL-CLAD (FOR PRINTED WIRING BOARDS), BASE MATERIAL GP, GLASS BASE, NON WOVEN, POLYTETRAFLUOROETHYLENE RESIN, FLAME RESISTANT, COPPER-CLAD

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the plastic sheets described herein shall consist of this specification and the latest issue of MIL-P-13949.

REQUIREMENTS:

Property values: See table I.

TABLE I. Property values.

Property to be tested	Test condition 1/	Thickness (inch) Under 0.020 copper types 2/		0.020 and over copper types 2/	
		A B E F	C D	A B E F	C D
I. Peel strength, minimum in lbs-per-inch width					
A. Under 1/2-oz/ft^2 copper					
As received	A	---	5.0	---	6.0
After thermal stress	A	---	5.0	---	6.0
At elevated temperature	150°C	---	4.0	---	5.0
After exposure to processing solutions	A	---	4.0	---	6.0
B. 1/2-oz/ft^2 copper					
As received	A	4.0	5.0	4.0	6.0
After thermal stress	A	4.0	5.0	4.0	6.0
At elevated temperature	150°C	3.0	4.0	3.0	5.0
After exposure to processing solutions	A	3.0	4.0	3.0	5.0
C. 3/4-oz/ft^2 copper					
As received	A	4.0	6.0	4.0	7.0
After thermal stress	A	4.0	6.0	4.0	7.0
At elevated temperature	150°C	3.0	5.0	3.0	6.0
After exposure to processing solutions	A	3.0	5.0	3.0	6.0
D. 1-oz/ft^2 copper					
As received	A	5.0	7.0	5.0	8.0
After thermal stress	A	5.0	7.0	5.0	8.0
At elevated temperature	150°C	4.0	5.0	4.0	6.0
After exposure to processing solutions	A	4.0	5.0	4.0	6.0

See footnotes at end of table.

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After moisture resistance At elevated temperature	F E-24/150	All All	10^4 10^3
4. Water absorption, maximum percent	E-1/105 +des +D-24/23	(inch) 0.031 0.062 0.093 0.125 0.250	 0.30 0.20 0.20 0.20 0.15
5. Dielectric breakdown (parallel to laminations), minimum kV, step-by-step	D-48/50+ D-0.5/23	Applicable only for 0.020 inch and over.	30
6. Permittivity at 1 MHz max at 50 MHz max	C-24/23/50	All 3/	2.4
7. Loss tangent at 1 MHz max. at 50 MHz max.	C-24/23/50	All 3/	0.0010
8 Q (resonance), minimum at 1 MHz at 50 MHz at 100 MHz	N/A	N/A	N/A

See footnotes at end of table.

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TABLE I. Property values - Continued.

Property to be tested	Test condition 1/	Thickness	Values required
9. Flexural strength, minimum thousand lbs-per-square inch			
A. Lengthwise	A	Applicable only for 0.031 inch and over.	8.0
B. Crosswise	A		6.0
C. Lengthwise at elevated temperature	N/A	N/A	N/A
10. Arc resistance, minimum seconds	D-48/50+ D-0.5/23	0.031 and over under 0.031	180 120
11. Flammability, maximum burntime, 4/ seconds, each flame application	N/A	N/A	N/A

1/ See specimen conditioning requirements, conditioning tolerances and testing information table XXVI of MIL-P-13949.
2/ Copper types (see 1.2.1.1.3).
3/ Materials other than 0.030 and 0.060 inch thick are not testable at 10 GHz.
4/ PTFE materials will not burn in small scale flame tests.

Thickness: Cut sheets less than 18 inches by 16 inches shall meet the applicable thickness tolerances in 100 percent of the area of the sheet.

Marking: Internal marking not applicable.

NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities.
Army - AR, MI
Navy - AS, OS, SH
Air Force - 11, 85
NSA - S5
DLA - ES

User activities:
Army - AT, ME
Navy - MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5999-0161-06)

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